

Advanced Engineering Mathematics

Partial Differential Equations

Professor:

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Competence	Outcome
Analyze the behavior of dynamic systems with distributed parameters through modeling with partial differential equations.	• Model dynamic systems with distributed parameters using mass, energy, and population balances. • Model dynamic systems with partial differential equations in one and multiple dimensions. • Solve standard cases of partial differential equations. • Implement numerical strategies for solving dynamic systems described by partial differential equations.

1 Theoretical framework: one-dimensional heat conduction in an isolated rod

The experimental setup consists of a one-dimensional rod whose lateral surface is thermally insulated, while the temperatures at the two boundaries are maintained at different constant values. Initially, the rod is assumed to have a uniform temperature distribution. The objective of the experiment is to observe the transient heat diffusion process and compare the measured temperature evolution with the theoretical predictions derived from the heat equation and numerical simulation.

The theoretical framework of this laboratory is based on the formulation of the one-dimensional heat conduction equation presented in the textbook *Applied Partial Differential Equations with Fourier Series and Boundary Value Problems* by Richard Haberman.

1.1 Heat transfer by conduction

Heat transfer in solids occurs primarily through conduction, a mechanism in which thermal energy is transferred between neighboring molecules due to microscopic interactions. In the present experiment, convection effects are neglected because the rod is thermally insulated along its lateral surface, allowing heat flow only along the axial direction of the rod.

Under these assumptions, the rod can be modeled as a one-dimensional domain extending from $x = 0$ to $x = L$, where x represents the spatial coordinate along the rod and L is the rod length. The assumption of one-dimensional conduction implies that temperature gradients across the cross section are negligible.

1.2 Conservation of thermal energy

The derivation of the governing equation begins with the principle of conservation of thermal energy. Considering a differential element of the rod between x and $x + \Delta x$, the variation of thermal energy inside the element equals the net heat flux entering the element plus any internal heat generation.

For the experimental configuration considered here:

- the rod is homogeneous,
- there are no internal heat sources,
- the cross-sectional area is constant,
- lateral losses are neglected.

Under these conditions, conservation of energy leads to the classical one-dimensional heat equation.

1.3 Fourier's law of heat conduction

The relationship between heat flux and temperature gradient is established through Fourier's law of heat conduction:

$$\phi = -K_0 \frac{\partial u}{\partial x}, \quad (1)$$

where ϕ is the heat flux, K_0 is the thermal conductivity, and $u(x, t)$ is the temperature field.

The negative sign indicates that heat flows from regions of higher temperature toward regions of lower temperature.

1.4 Derivation of the heat equation

Combining the conservation of energy with Fourier's law yields the general heat conduction equation:

$$c\rho \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(K_0 \frac{\partial u}{\partial x} \right) + Q, \quad (2)$$

where c is the specific heat, ρ is the density, and Q is the internal heat generation term.

For a uniform rod with constant thermal properties and no internal heat generation ($Q = 0$), the equation reduces to:

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad (3)$$

where the thermal diffusivity is defined as

$$k = \frac{K_0}{c\rho}. \quad (4)$$

This equation describes the transient diffusion of temperature inside the rod.

1.5 Initial condition

The experiment starts from a uniform initial temperature distribution. Therefore, the initial condition can be expressed as

$$u(x, 0) = T_i, \quad (5)$$

where T_i is the initial temperature of the rod before the boundary temperatures are imposed. The initial condition represents the thermal state of the rod at $t = 0$.

1.6 Boundary conditions

The experimental setup imposes constant temperatures at both ends of the rod. Therefore, Dirichlet boundary conditions are used:

$$u(0, t) = T_1, \text{ and } u(L, t) = T_2. \quad (6)$$

where T_1 is the imposed temperature at the left boundary and T_2 is the imposed temperature at the right boundary.

Because $T_1 \neq T_2$, a temperature gradient is established along the rod, generating heat flow from the hotter boundary toward the colder boundary.

1.7 Equilibrium temperature distribution

As time progresses, the transient thermal behavior approaches a steady-state condition in which the temperature no longer changes with time. In equilibrium:

$$\frac{\partial u}{\partial t} = 0. \quad (7)$$

Consequently, the governing equation reduces to

$$\frac{d^2 u}{dx^2} = 0, \quad (8)$$

whose solution is a linear temperature distribution:

$$u(x) = T_1 + \frac{T_2 - T_1}{L}x. \quad (9)$$

This steady-state solution represents the final temperature profile expected experimentally after sufficient time has elapsed.

1.8 Physical interpretation of the experiment

The experiment illustrates the diffusion nature of heat transfer. Initially, the rod is at a uniform temperature T_i . Once the boundary temperatures T_1 and T_2 are imposed, thermal energy begins to redistribute through conduction.

The transient response depends on:

- the thermal diffusivity k ,
- the rod length L ,
- the imposed temperature difference,
- the initial temperature distribution.

During the transient regime, the temperature profile evolves continuously until the steady-state linear distribution is reached.

The laboratory measurements allow students to:

1. Observe transient heat diffusion experimentally,
2. Validate Fourier's law,
3. Compare experimental data with analytical and numerical PDE solutions,
4. Understand the role of boundary and initial conditions,
5. Analyze the influence of thermal diffusivity on system dynamics.

The one-dimensional heat equation is one of the fundamental partial differential equations in engineering and physics. Similar diffusion models appear in:

- thermal systems,
- mass diffusion,

- electrical transmission lines,
- fluid transport,
- control systems,
- biomedical heat transfer.

Therefore, this laboratory provides a practical introduction to distributed parameter systems and partial differential equation modeling, both of which are essential in advanced engineering analysis.

2 Simulation of the one-dimensional heat equation

In this section, students must perform a numerical simulation of the analytical solution, studying the transient heat conduction in a one-dimensional copper alloy rod subjected to different temperatures at its boundaries. The objective is to analyze the transient thermal behavior of the rod and compare the evolution of the temperature distribution over time with the theoretical analytical solution of the heat equation.

2.1 Problem description

Consider a homogeneous copper alloy rod of length $L = 80$ cm with thermally insulated lateral surfaces, such that heat transfer occurs only along the axial direction. The temperature distribution inside the rod is governed by the one-dimensional heat equation

$$\frac{\partial u(x, t)}{\partial t} = k \frac{\partial^2 u(x, t)}{\partial x^2}, \quad (10)$$

where $u(x, t)$ is the temperature distribution, k is the thermal diffusivity of the copper alloy (use a commercial value as an approximation), $x \in [0, L]$ is the spatial coordinate, and $t \geq 0$ is time.

The rod initially starts at a constant temperature $u(x, 0) = T_i$. The initial temperature T_i may be selected as an approximation of the room temperature.

The boundary conditions are prescribed temperatures at both ends of the rod:

$$u(0, t) = T_1, \text{ and } u(L, t) = T_2.$$

Students must select the boundary temperatures within the following ranges:

$$T_1 \in [4, 15]^\circ C, \text{ and } T_2 \in [30, 80]^\circ C.$$

2.2 Objectives

The objectives of this activity are:

1. Derive the analytical solution of the one-dimensional transient heat equation.
2. Analyze the transient response of the rod temperature distribution.
3. Study the influence of boundary conditions on the heat diffusion process.
4. Compare the transient and steady-state temperature distributions.
5. Visualize the temperature evolution using graphical representations.

2.3 Simulation requirements

Students must derive the analytical solution of the heat equation using the method of separation of variables. Then, implement a simulation of the analytical solution using computational tools of Julia.

The simulation must include:

1. The definition of the rod parameters:
 - rod length L ,
 - thermal diffusivity k ,
 - initial temperature T_i ,
 - boundary temperatures T_1 and T_2 .
2. Calculation of the Fourier coefficients.
3. Approximation of the infinite series using a sufficient number of terms.
4. Evaluation of the temperature distribution for multiple time instants.
5. Visualization of the transient response.

2.4 Required graphical results

Students must generate a graphical representation of the three-dimensional surface plot of temperature evolution:

$$u(x, t)$$

as a function of both space and time.

Additionally, students must show the temporal evolution of the temperature at discrete positions along the rod with a distance of 10 cm between them.

2.5 Analysis questions

Students must discuss the following aspects:

1. How does the temperature profile evolve with time?
2. How does the thermal diffusivity affect the transient response?
3. Why does the solution eventually converge to the steady-state distribution?
4. What is the physical interpretation of the exponential decay terms?
5. How many Fourier series terms are required to obtain a sufficiently accurate approximation?

3 Experimental validation of the one-dimensional heat equation

After completing the analytical derivation and computational simulation of the transient heat conduction problem, students must design a physical experiment equivalent to the mathematical model described previously. The objective of this activity is to compare the analytical solution, the numerical simulation, and the experimental measurements obtained from a real thermal system.

3.1 Experimental objectives

The objectives of the experimental activity are:

1. Reproduce experimentally the one-dimensional transient heat conduction problem.
2. Validate the analytical solution of the heat equation.
3. Compare theoretical, simulated, and experimental temperature distributions.
4. Analyze discrepancies between mathematical models and physical systems.
5. Study the practical limitations of heat-transfer modeling.

3.2 Experimental setup requirements

Students must define and justify all experimental conditions required to reproduce the same conditions used in the simulation. The experimental system includes:

- A copper-alloy (unknown) rod of length L .

- Thermal insulation along the lateral surface of the rod.
- Two thermal elements capable of maintaining fixed temperatures at both ends.
- Multiple temperature sensors distributed along the rod (10 cm).
- A data acquisition system for recording temperature evolution over time (sampling time = 0.1 s).

The experiment emulates the following assumptions used in the mathematical model:

1. One-dimensional heat transfer.
2. Constant thermal properties.
3. No internal heat generation.
4. Constant boundary temperatures.
5. Uniform initial temperature.

3.3 Definition of experimental conditions

Students must explicitly define the following experimental parameters:

1. Rod material properties:
 - thermal conductivity,
 - density,
 - specific heat,
 - thermal diffusivity.
 - total experiment duration = 5 minutes,
2. Geometrical properties:
 - rod length = 80 cm,
 - rod diameter = 3 mm,
 - sensors locations = 10 cm.

3. Initial condition:

$$u(x, 0) = T_i.$$

4. Boundary conditions:

$$u(0, t) = T_1 \in [4, 15]^\circ C,$$

$$u(L, t) = T_2 \in [30, 80]^\circ C.$$

3.4 Experimental setup

1. The rod has thermal insulation on its lateral surface.
2. The rod is placed in a thermal environment until a uniform initial temperature T_i is reached.
3. Heating and cooling elements are placed at both ends of the rod at $t=15$ s.
4. Boundary temperatures T_1 and T_2 are set at values employed in the simulation.
5. The temperature evolution at all sensors is recorded with a sampling period of $T_s=0.1$ s.
6. Data acquisition is recorded for 5 minutes.

3.5 Comparison between theory and experiment

Students must compare the analytical solution and experimental measurements.

The comparison must include: temperature profiles at different time instants, temporal evolution at selected sensor locations, steady-state temperature distribution, and transient response characteristics.

Students must generate comparative plots showing:

1. Analytical solution versus experimental measurements.
2. Error evolution as a function of time and position.
3. Three-dimensional temperature evolution plots.

3.6 Error analysis

Students must quantify the experimental error using appropriate metrics such as:

$$\text{Mean Absolute Error (MAE)} = \frac{1}{N} \sum_{i=1}^N |u_{\text{exp}} - u_{\text{theory}}|, \quad (11)$$

or

$$\text{Root Mean Square Error (RMSE)} = \sqrt{\frac{1}{N} \sum_{i=1}^N (u_{\text{exp}} - u_{\text{theory}})^2}. \quad (12)$$

Analyze possible sources of discrepancy between the analytical model and the physical experiment.

3.7 Discussion and conclusions

Students must discuss:

1. Whether the experimental results validate the heat equation model.
2. The quality of agreement between:
 - numerical simulation,
 - physical experiment.
3. The influence of nonideal experimental conditions.
4. The limitations of the one-dimensional heat conduction assumption.
5. The importance of boundary and initial conditions in distributed parameter systems.
6. The physical interpretation of transient heat diffusion.

Finally, students must conclude whether the heat equation provides an adequate mathematical representation of the experimental thermal system and explain the origin of any observed deviations.

4 Instruction

1. Define T_1 and T_2 within the limits.
2. Request your experimental data to the teacher assistant at the email: `alejandro.arroyaver@udea.edu.co` using "Asunto: Experimento Matemáticas Especiales". Include a copy email to: `jaime.arcos@udea.edu.co`.
3. Experiments must be requested before may/28th/2026 at 2:00 pm. **No experiments will be processed after the deadline.**
4. Requests will be attended to in order of arrival.
5. Work in groups of minimum three students and maximum four.
6. **Submit a file with five slides of your results and analysis.**